

Khachatur Pepanyan

Senior AI Engineer

Email: pepaniank@gmail.com Telegram: [yasinyan23](https://t.me/yasinyan23) LinkedIn: [yasinyan](https://www.linkedin.com/in/yasinyan)

SUMMARY

Senior AI Engineer and Lead Software Engineer with 8+ years of software engineering experience, including 4+ years building and productionizing AI systems across computer vision, 3D human motion reconstruction, LLMs, RAG, tool-calling agents, document intelligence, model evaluation, and inference optimization. Experienced across the full AI lifecycle — from model selection, fine-tuning, and evaluation to inference, model serving, distributed processing, observability, and integration into production products.

Built AI systems at scale in sports technology and fintech, backed by a strong enterprise software engineering background. Production computer-vision systems run on 30 inference workers on an 80-vCPU GCP instance, sustaining 20k+ plays/day of processing throughput; ~68k historical plays were reprocessed in ~3 days, generating 20M+ biomechanics samples for analytics. Previously led credit-decisioning platforms processing 50k–70k requests/day, increasing throughput by ~45%, reducing decision time by ~80%, and cutting manual reviews by 60%+. Built LLM-powered features on top of these platforms, reducing review handling time by 52%, achieving 96% document field-level accuracy, and reducing AI cost per case by 45%. Strong production engineering background with Python, PyTorch, FastAPI, RabbitMQ, Celery, PostgreSQL, Docker, GCP, CI/CD, model serving, and AI evaluation. Experienced in taking research prototypes into reliable production systems, optimizing inference for latency and resource constraints, and leading teams of 5–7 engineers. Holds a Master's degree in Computer Engineering, providing a strong foundation in software architecture and systems engineering for building scalable AI products.

WORKING EXPERIENCE

FitWise AI

Feb 2026 – Present

AI Platform & Sports Technology Company. Develops an AI-powered platform for athlete video analysis, combining computer vision, 3D motion reconstruction, video processing, and backend product workflows to turn athlete footage into biomechanics insights.

PATENT / IP - CO-INVENTOR: System and Method for Automated Sprint Biomechanics Assessment of Athletes from Reconstructed Three-Dimensional Body Motion

Senior AI Software Engineer — AI Platform & Product

Own end-to-end production AI systems that turn match video into biomechanics insights, from computer-vision model selection and fine-tuning through PyTorch inference, 3D human body reconstruction, model serving, distributed processing, validation, and product integration. Built and scaled an asynchronous AI pipeline sustaining 20k+ plays/day of processing throughput across 30 inference workers on an 80-vCPU GCP instance, reprocessed ~68k historical plays in ~3 days, and transformed model output into 20M+ production biomechanics samples while delivering athlete insights within 3 minutes of video upload. Partner with biomechanics scientists, frontend, and QA to take research prototypes into reliable production systems.

- Productionized a multi-stage computer-vision pipeline for match video, covering player detection and tracking, per-frame 3D human body reconstruction with PyTorch and a parametric body model, gait-cycle segmentation aligned to toe-off, and 34 biomechanics metrics per cycle; delivers athlete insights within 3 minutes of video upload. Evaluated, selected, and fine-tuned computer-vision models for production inference, balancing model accuracy against latency and CPU/memory constraints; managed and versioned model weights and checkpoints across releases.
- Built preprocessing and postprocessing pipelines around model inference, including frame extraction, player cropping, keypoint filtering, temporal smoothing, gait-phase detection, and toe-off detection, improving the quality of downstream biomechanics metrics and overall pipeline throughput. Deployed PyTorch models as a production serving tier using long-lived inference workers that loaded model weights once and reused them across tasks; kept memory at ~2.4 GiB per worker on average, allowing 30 workers to run on a single 80 GB host, with Docker Compose, GitLab CI, and Google Artifact Registry.
- Profiled and optimized CPU inference at fleet scale, tracing a throughput bottleneck to PyTorch thread oversubscription (load average ~837 on 80 cores), tuning OMP/MKL and PyTorch intra-op thread pools for a 30–40% throughput gain, and moving workers to SSD persistent disks to eliminate I/O stalls under full production load. Scaled asynchronous AI inference to 30 Celery worker replicas on an 80-vCPU GCP instance using RabbitMQ, late acknowledgements, retries, and backoff; reprocessed the historical catalog of ~68k plays in ~3 days, sustaining 20k+ plays per day.
- Built production recovery and reprocessing mechanisms across the inference pipeline, including bounded connection pools between inference services, per-play processing status, failure recovery, and asynchronous task execution. Built AI output validation and QA workflows, including value-for-value comparison of the new per-cycle metrics path against the legacy implementation on production data with 0 mismatches, per-cycle completeness logging to expose undetected gait phases, and a human-in-the-loop workflow for annotators reviewing model output. Took biomechanics research prototypes into production with domain scientists, moving left/right asymmetry analysis from a browser prototype into backend services with cohort baselines and per-metric history endpoints while keeping domain formulas owned by the biomechanics experts. Transformed raw 3D motion inference output into queryable production data, storing up to 340 values per gait cycle (34 metrics across 5 gait phases and both legs) and 20M+ metric samples in PostgreSQL for player baselines, cohort comparisons, and left/right asymmetry analytics.
- Developed the end-to-end asynchronous video-processing workflow from video ingestion through computer-vision-based 3D motion reconstruction and biomechanics analysis delivery using Python, FastAPI, asynchronous processing, computer vision, and 3D reconstruction. Implemented backend services and product APIs connecting AI processing pipelines with application workflows, adding input validation, structured logging, distributed tracing, and automated tests for production observability and troubleshooting.
- Designed and implemented FastAPI authentication and session management with layered architecture, supporting login, token validation/refresh, current-user lookup, and logout. Modeled users, organizations, memberships, and sessions in PostgreSQL for organization-based access control, with 91 unit tests plus smoke tests. Developed APIs for games, jobs, plays, and runs, including QA workflows, job-status enrichment, and season-aware player numbering with database-backed uniqueness and asynchronous upserts.
- Maintained CI/CD for API, workers, PostgreSQL, RabbitMQ, and Flower using Docker Compose, Buildx, Docker-in-Docker, and Terraform; managed images in Google Artifact Registry and resolved registry authentication/permission issues blocking production deployments.

